

Integration WS 11

Evaluate Q1-9

1. $\int \frac{x^2+x+1}{x^2-x+1} dx$

2. $\int_0^1 \frac{(x+1)dx}{x^2+x+1}$

3. $\int_{-1}^0 \frac{2x+1}{x^2+6x+10} dx$

4. $\int_a^b \frac{x^3}{1-x} dx \quad (1 < a < b)$

5. $\int \frac{x}{x^4-1} dx$

6. $\int \frac{dx}{(x+2)^2(x^2+4)}$

7. $\int_1^2 \frac{dx}{x^2(x+2)}$

8. $\int_0^1 \frac{2x dx}{(x+2)(x^2+4)}$

9. $\int_0^{\frac{\pi}{2}} \frac{\cos \theta}{\sin^2 \theta + 5 \sin \theta + 6} d\theta$

10. Consider $I = \int_{-1}^1 \frac{x^2 e^x}{e^x+1} dx$ and $J = \int_{-1}^1 \frac{x^2}{e^x+1} dx$

a. Use the substitution $u = -x$ in I to show that $I = J$.

b. Hence evaluate I .

11. Consider the integral $I_n = \int_0^1 \frac{1}{(1+x^2)^n} dx$ for $n \geq 1$.

a. Use integration by parts to show that

$$I_n = \frac{1}{2n} + 2n \int_0^1 \frac{x^2}{(1+x^2)^{n+1}} dx$$

b. Find the values of A and B for which

$$\frac{A}{(1+x^2)^n} + \frac{B}{(1+x^2)^{n+1}} = \frac{x^2}{(1+x^2)^{n+1}}$$

c. Show that $I_{n+1} = \frac{1}{n2^{n+1}} + \frac{2n-1}{2n} I_n$

d. Hence obtain the exact value of

$$\int_0^1 \frac{1}{(1+x^2)^3} dx$$

12. Suppose that $f(x)$ is an even function and $a > 0$ is a constant.

a. Prove that $\int_{-a}^a \frac{f(x)}{1+e^x} dx = \int_0^a f(x) dx$

b. Hence, evaluate $\int_0^1 \frac{(1-e^x)^2}{(1+e^x)^3} dx$

15. a. Express $\frac{1}{1+x+x^2+x^3}$ in the form $\frac{A}{1+x} + \frac{Bx+C}{1+x^2}$,

where A, B and C are rational numbers.

b. Hence evaluate $\int_0^1 \frac{dx}{1+x+x^2+x^3}$.

16. a. Given that $I_n = \int_0^{\frac{\pi}{2}} \theta \sin^n \theta d\theta$, prove $I_n = \frac{n-1}{n} I_{n-2} + \frac{1}{n^2}$ for $n \geq 2$.

b. Hence show that $\int_0^{\frac{\pi}{2}} \theta \sin^5 \theta d\theta = \frac{149}{225}$.

17. a. Use the substitution $t = \tan \frac{x}{2}$ to prove that

$$\int_0^{\frac{\pi}{2}} \frac{dx}{2+\sin x} = \frac{\pi}{3\sqrt{3}}.$$

b. Show that $\int_0^{2a} f(x) dx = \int_0^a \{f(x) + f(2a-x)\} dx$

c. Hence evaluate $\int_0^{\pi} \frac{x}{2+\sin x} dx$.

16. Given $I_n = \int_1^e (\ln x)^n dx$, for $n = 0, 1, 2, \dots$

a. Show that $I_n = e - nI_{n-1}$, for $n = 1, 2, 3, \dots$

b. Let $J_n = \frac{I_n}{n!}$, show that $\frac{1}{e} (1 + J_{10}) = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + \frac{1}{10!}$

c. Given that $\sum_{r=0}^n \frac{(-1)^r}{r!} = \frac{1}{e} [1 + (-1)^n J_n]$, for $n = 0, 1, 2, \dots$. Deduce the sum to infinity of the series:

$$\sum_{r=0}^{\infty} \frac{(-1)^r}{r!} = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots$$

Justify your answer.

17. Let $I_n = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \operatorname{cosec}^n x dx$ where n is a positive integer.

a. Show $I_n = \frac{2^{n-2}\sqrt{3}}{n-1} + \frac{n-2}{n-1} I_{n-2}$

b. Evaluate I_3 .

20. a. Show that $(1 - \sqrt{x})^{n-1} \sqrt{x} = (1 - \sqrt{x})^{n-1} - (1 - \sqrt{x})^n$

b. If $I_n = \int_0^1 (1 - \sqrt{x})^n dx$ for $n \geq 0$ show

$$\text{that } I_n = \frac{n}{n+2} I_{n-1} \text{ for } n \geq 1$$

c. Hence find I_3

19. Let $I_n = \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2nx}{\sin 2x} dx$ for $n = 1, 2, 3 \dots$

a. Evaluate I_1

b. Using the fact that $\cos S - \cos T =$

$$-2 \sin \frac{S+T}{2} \sin \frac{S-T}{2}, \text{ show that for } r \geq 1,$$

$$I_{2r+1} - I_{2r-1} = \frac{1 - (-1)^r}{2r}$$

c. Hence evaluate I_7

20. Evaluate these by first multiplying or dividing

by an appropriate factor.

a. $\int_1^2 \frac{e^{2x} + 1}{e^{2x} - 1} dx$

b. $\int_0^1 \frac{e^x}{e^x + e^{-x}} dx$

c. $\int_1^{\sqrt{3}} \frac{2 + \frac{1}{x}}{x + \frac{1}{x}} dx$

21. Find

a. $\int_1^e \sin(\ln x) dx$

b. $\int_1^e \cos(\ln x) dx$

22. If $\ln \int_0^{\frac{\pi}{2}} \frac{\sin(2n+1)\theta}{\sin \theta} d\theta$, show that $I_n - I_{n-1} = 0$

for $n \geq 1$. Hence find the value of I_n for $n \geq$

0.

Answer

1. $x + \ln(x^2 - x + 1) + \frac{1}{\sqrt{3}} \tan^{-1} \frac{2x-1}{\sqrt{3}}$

2. $\frac{\ln 3}{2} + \frac{\pi}{6\sqrt{3}}$

3. $\ln 2 - 5 \tan^{-1} \left(\frac{1}{7} \right)$

4. $\frac{1}{3}(a^3 - b^3) + \frac{1}{2}(a^2 - b^2) + (a - b) + \ln \left(\frac{a-1}{b-1} \right)$

5. $\frac{1}{4} \ln(x^2 - 1) - \frac{1}{4} \ln(x^2 + 1)$

6. $\frac{1}{16} \ln(x+2) - \frac{1}{8(x+2)} - \frac{1}{32} \ln(x^2 + 4)$

7. $\frac{1}{4} + \frac{1}{4} \ln \left(\frac{2}{3} \right)$

8. $-\frac{1}{2} \ln \left(\frac{3}{2} \right) + \frac{1}{4} \ln \left(\frac{5}{4} \right) + \frac{1}{2} \tan^{-1} \frac{1}{2}$

9. $\ln \left(\frac{9}{8} \right)$

10. $\frac{1}{3}$

11. b. $A = 1, B = -1$; d. $\frac{1}{4} + \frac{3\pi}{32}$

12. a. $A = \frac{1}{2}, B = -\frac{1}{2}, C = \frac{1}{2}$,
 $\frac{1}{2(1+x)} + \frac{-x+1}{2(1+x^2)}$; b. $\frac{\ln 2}{4} + \frac{\pi}{8}$

13.

14. c. $\frac{\pi^2}{3\sqrt{3}}$

15. c. $\frac{1}{2}$

16. b. $I_3 = \sqrt{3} + \frac{1}{2} \ln(2 + \sqrt{3})$

17. (20)

18. a. $\ln \sqrt{2} = \frac{1}{2} \ln 2$; c. $\frac{4}{3} + \frac{1}{2} \ln 2$

23. Let $I = \int_0^{\frac{\pi}{2}} \frac{1}{1 + \cos \alpha \sin x} dx$, where $0 < \alpha < \frac{\pi}{2}$.

a. Show that $\frac{1 + \cos \alpha}{\sin \alpha} = \tan \left(\frac{\pi}{2} - \frac{\alpha}{2} \right)$.

b. Use the substitution $t = \tan \frac{x}{2}$ to show

$$\text{that } \int_0^1 \frac{2}{(t + \cos \alpha)^2 + \sin^2 \alpha} dt$$

c. Use the further substitution $t + \cos \alpha = \sin \alpha \tan u$ to show that $I = \frac{\alpha}{\sin \alpha}$.

24. a. Use a suitable double angle formula to

$$\text{show that } \tan \frac{\pi}{8} = \sqrt{2} - 1.$$

b. Find $\cos 4\theta$ in terms of $\cos \theta$.

c. Let $I = \int_{-1}^1 \frac{1}{\sqrt{1+x}\sqrt{1-x}+2} dx$.

i. Show, by using the substitution $x =$

$$\sin 4\theta, \text{ that } I = \int_0^{\frac{\pi}{8}} \frac{2 \cos 4\theta}{\cos^2 \theta} d\theta.$$

ii. Hence find the exact value of I .

25. a. Find the values of A, B and C such that

$$\frac{1}{x^3 + x^2 + x + 1} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$$

b. Find and simplify $I(N) = \int_0^N \frac{1}{x^3 + x^2 + x + 1} dx$

c. Hence evaluate $\int_0^\infty \frac{dx}{x^3 + x^2 + x + 1}$.

19. a. $\ln(e + e^{-1}) + C$; b. $\frac{1}{2} \log \left(\frac{e^2 + 1}{2} \right)$;

c. $\frac{\pi}{12} + \log 2$

20. a. $\frac{1}{2} e(\sin 1 - \cos 1) + \frac{1}{2}$;

b. $\frac{1}{2} e(\sin 1 - \cos 1) - \frac{1}{2}$

21. a. 1; b. $\frac{1}{3} \ln 2$; c. $\frac{\pi}{4}$

22. $\frac{\pi}{2}$

23. proving

24. a. $\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$; c. ii. $4\sqrt{2} - \pi - 2$

25. a. $A = \frac{1}{2}, B = -\frac{1}{2}, C = \frac{1}{2}$;

b. $\frac{1}{4} \left(\log \left(\frac{(N+1)^2}{N^2+1} \right) + 2 \tan^{-1} N \right)$; c. $\frac{\pi}{4}$